
AN ENGINEERING DECISION

A queue is not **extra capacity.**

Buffer the burst. Budget the recovery.

01

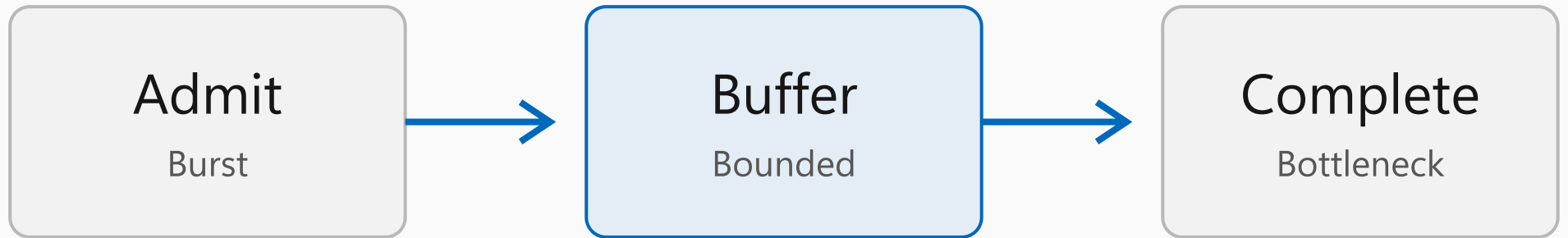
Capacity and waiting

THE ILLUSION

Fast acceptance.
Unchanged capacity.

The wait moved. The work did not.

A waiting room.
Not a wider exit.



RECOVERY

Only spare capacity drains the backlog.

- | Completions must outrun new arrivals.
- | Full utilization leaves nothing for recovery.

02

The hidden cost of acceptance

Watch the oldest unfinished promise.

- | Queue depth counts work.
- | Job age exposes delay.
- | Include in-flight jobs.

Retries are **more arrivals.**

- | One identity per logical job.
- | One finite retry budget.

03

A bounded promise

ADMISSION

Bound the promise, not just the queue.

- | Useful completion deadline.
- | Hard storage limit.
- | Honest rejection.

Every job needs a **visible ending**.

- | Completed.
- | Failed.
- | Explicitly expired.

THE DECISION TEST

Prove recovery **under continuing load.**

- | Burst. Slow dependency. Worker restart.
- | Account for every accepted job.

THE KEY MESSAGE

Buffer the burst. **Budget the recovery.**

A queue buys time, not capacity.